TECHWIZ ACADEMY

# NETWORKING BASICS

IP Addresses | Ports | Protocols | OSI Model

| | |
|---|---|
| **COURSE** | Cyber Security & Ethical Hacking Internship |
| **ASSIGNMENT** | Week 1 · Task 2 — Networking Basics |
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| **SUBMISSION DATE** | 14 November 2026 |
| **MAXIMUM MARKS** | 10 |

A beginner-friendly practical report about basic computer networking.

# 1. Introduction

Networking means connecting computers and other devices so they can communicate and share information. Networking knowledge is important in ethical hacking because it helps us understand how devices, websites, and services communicate.

# 2. IP Address and MAC Address

## IP Address

An IP address is a logical address used to identify a device on a network. It helps data reach the correct device.

**IPv4:** Uses 32 bits and is written in four numbers separated by dots. Example: 192.168.1.10. Each number is from 0 to 255.

**IPv6:** Uses 128 bits and is written in hexadecimal groups separated by colons. Example: 2001:db8::1. IPv6 provides a much larger number of addresses than IPv4.

Private IP addresses are used inside local networks. Public IP addresses are used to communicate over the internet. A private IP is not directly reachable from the public internet in the normal way.

## MAC Address

A MAC address is a hardware address assigned to a network interface. It is mainly used for communication inside a local network. Example format: 00:1A:2B:3C:4D:5E.

**IP vs MAC:** An IP address helps route data between networks, while a MAC address helps deliver data on the local network.

# 3. Practical Work — Find Your Private IP

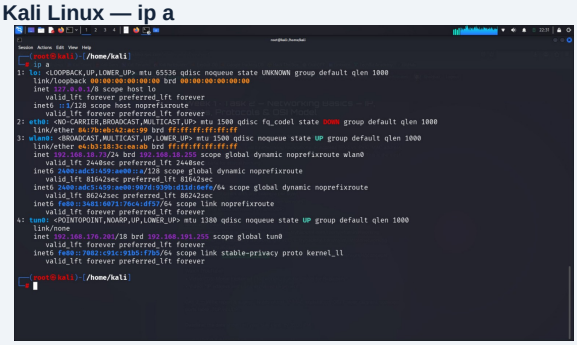
On Linux, open the Terminal and run: **ip a**. Look for the active network interface (for example, wlan0 or eth0) and find the line starting with **inet**. An address such as 192.168.1.18/24 is an example of a private IPv4 address. Do not copy this example as your own address.

**My private IPv4 address: 192.168.18.73/24 (Kali Linux, wlan0)**

You can also use **hostname -I** to display assigned IP addresses. For Windows, use **ipconfig** in Command Prompt and check the IPv4 Address field.

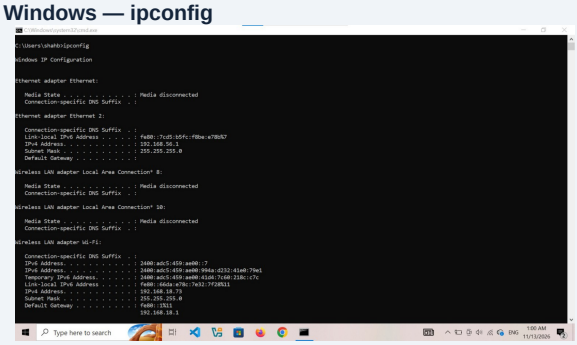
**Practical evidence:** Screenshots below show the actual command output from Kali Linux and Windows.

**Kali Linux — ip a**



192.168.18.73/24

**Windows — ipconfig**



192.168.18.73

# 4. The 7 Layers of the OSI Model

The OSI model explains network communication in seven layers. Each layer has a different job.

| Layer | Name | Main job | Example |
|---|---|---|---|
| 7 | Application | Provides network services to user applications. | HTTP, DNS |
| 6 | Presentation | Formats, encrypts, and compresses data. | TLS encryption, JPEG |
| 5 | Session | Starts, manages, and ends communication sessions. | Session management |
| 4 | Transport | Delivers data between applications; may provide reliability. | TCP, UDP |
| 3 | Network | Uses logical addresses and selects a route. | IP, routers |
| 2 | Data Link | Transfers frames on the local network using MAC addresses. | Ethernet, switches |
| 1 | Physical | Sends raw bits through cables, radio, or other media. | Fiber cable, Wi-Fi radio |

**Easy way to remember:** Application, Presentation, Session, Transport, Network, Data Link, Physical.

# 5. Common Network Ports and Services

A port is a number used by a device to direct network traffic to the correct service or application. TCP and UDP use port numbers.

| Port | Service | Protocol | Common use |
|---|---|---|---|
| 20 | FTP Data | TCP | Transfers file data in active FTP |
| 21 | FTP Control | TCP | Controls an FTP session |
| 22 | SSH | TCP | Secure remote login and command access |
| 23 | Telnet | TCP | Remote login; data is not encrypted |
| 25 | SMTP | TCP | Sends email between mail servers |
| 53 | DNS | TCP/UDP | Finds IP addresses for domain names |
| 80 | HTTP | TCP | Web traffic without TLS encryption |
| 110 | POP3 | TCP | Retrieves email from a mail server |
| 143 | IMAP | TCP | Accesses and manages email on a server |
| 443 | HTTPS | TCP | Encrypted web traffic using TLS |

## 6. TCP vs UDP

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are transport-layer protocols. They are used to send data between devices, but they work differently.

| TCP | UDP |
|---|---|
| Connection-oriented: it establishes a connection before sending data. | Connectionless: it sends data without setting up a connection first. |
| Reliable: it uses acknowledgements and retransmission to help deliver data. | No delivery guarantee: it does not normally retransmit lost data. |
| Usually has more overhead and can be slower. | Usually has less overhead and can be faster. |
| **Examples:** HTTPS, SSH, email. | **Examples:** DNS queries, live streaming, online gaming (depending on the application). |

## 7. TryHackMe Learning Activities

The following free rooms are part of the assignment. Complete each room using the exact links provided by the academy and add your completion screenshots to this report.

**COMPLETION EVIDENCE**

What is Networking? — Screenshot on page 5

Introductory Networking — Screenshot on page 5

Networking Concepts — Screenshot on page 6

## 8. Conclusion

In this task, I learned the basic ideas of computer networking. I learned how IP and MAC addresses identify devices, how ports identify network services, and how TCP and UDP send data. I also studied the seven layers of the OSI model. These topics are important for understanding networks and performing ethical hacking in a legal lab environment.

## 9. Practical Evidence Checklist

Before submitting, add your own evidence:

1. Screenshot showing your computer's actual private IP address.
2. Screenshot of each completed TryHackMe room (three rooms).
3. Check that your name, course, and submission date are correct.

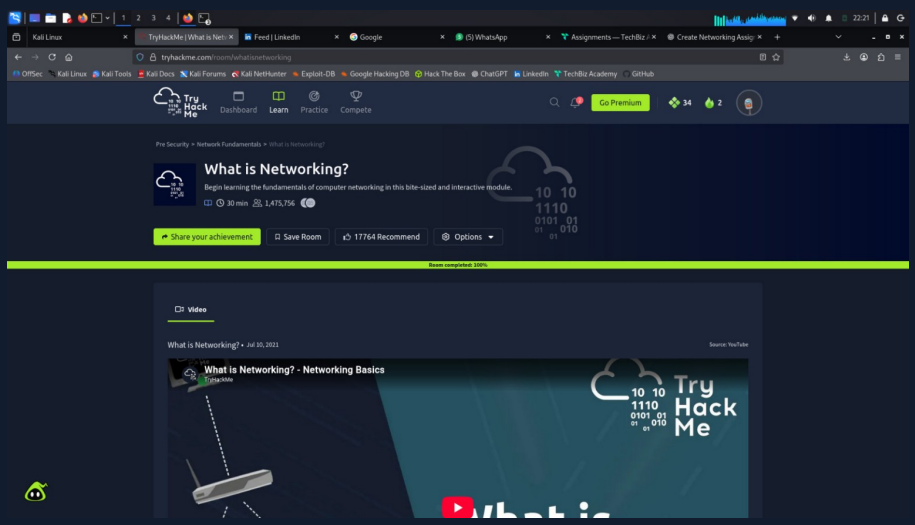
**Note:** The private IP field and screenshots are left for your actual practical results. Do not claim a room is completed until you have completed it and added proof.

# TRYHACKME COMPLETION EVIDENCE

Verified room completion screenshots supplied for Week 1 · Task 2

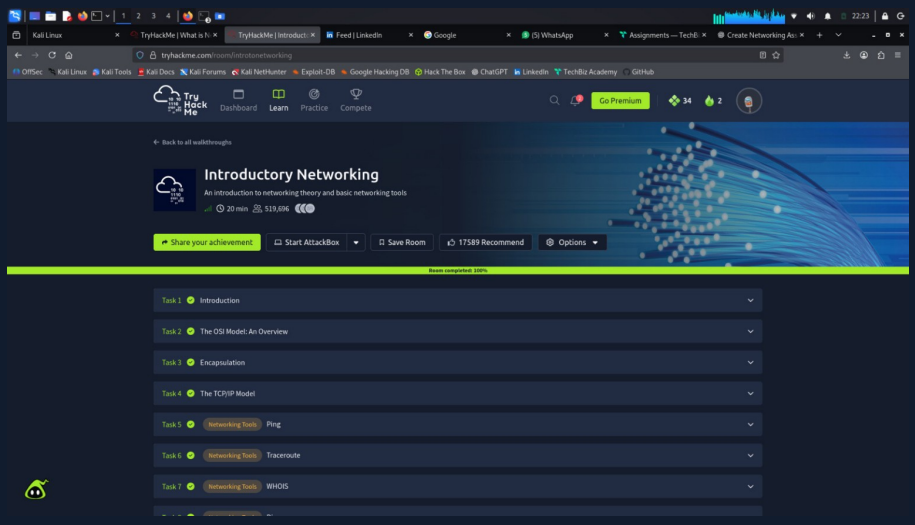
## What is Networking?

tryhackme.com/room/whatisnetworking



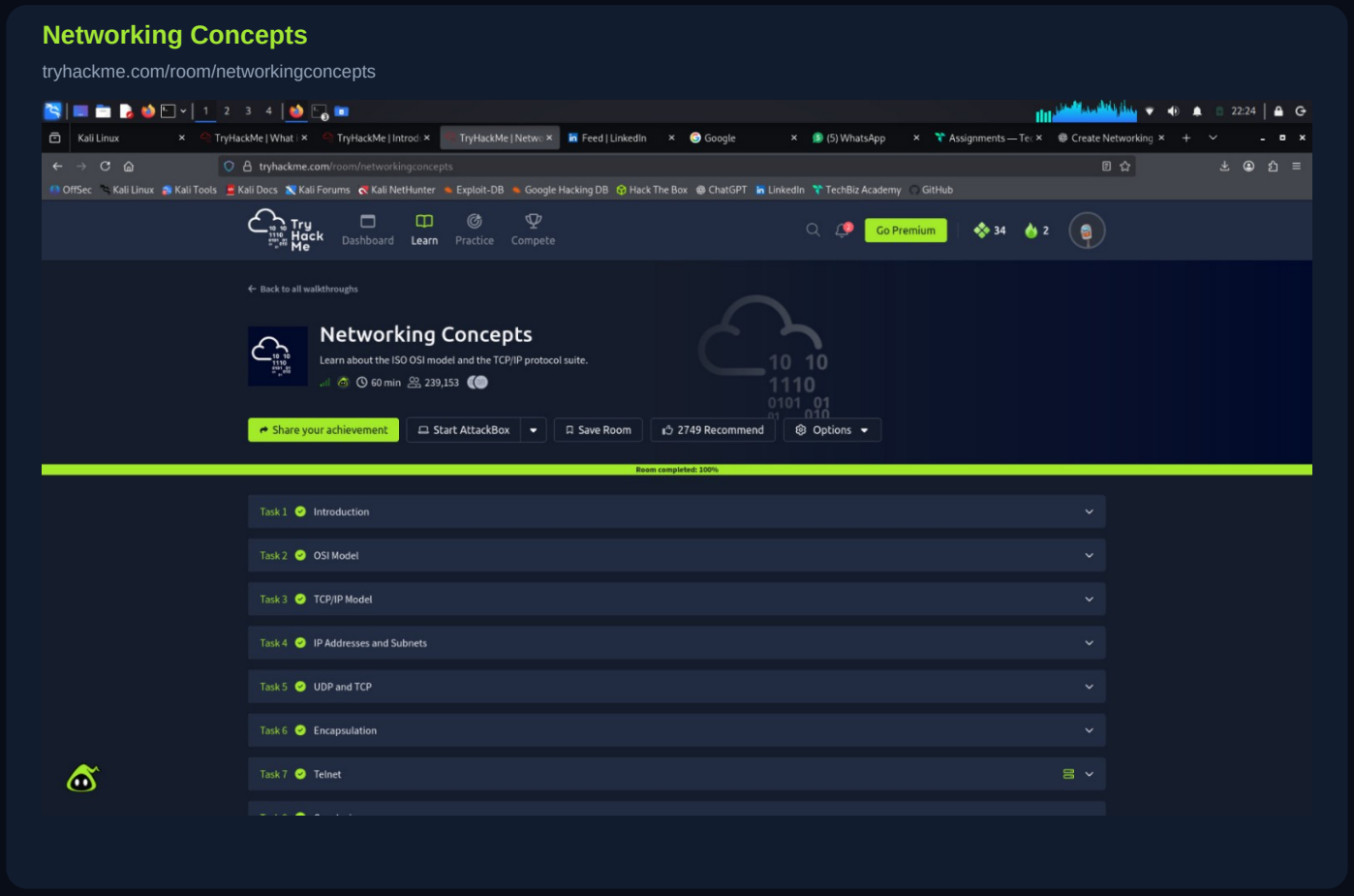

## Introductory Networking

tryhackme.com/room/introtonetworking



# TRYHACKME COMPLETION EVIDENCE

Networking Concepts room completion proof

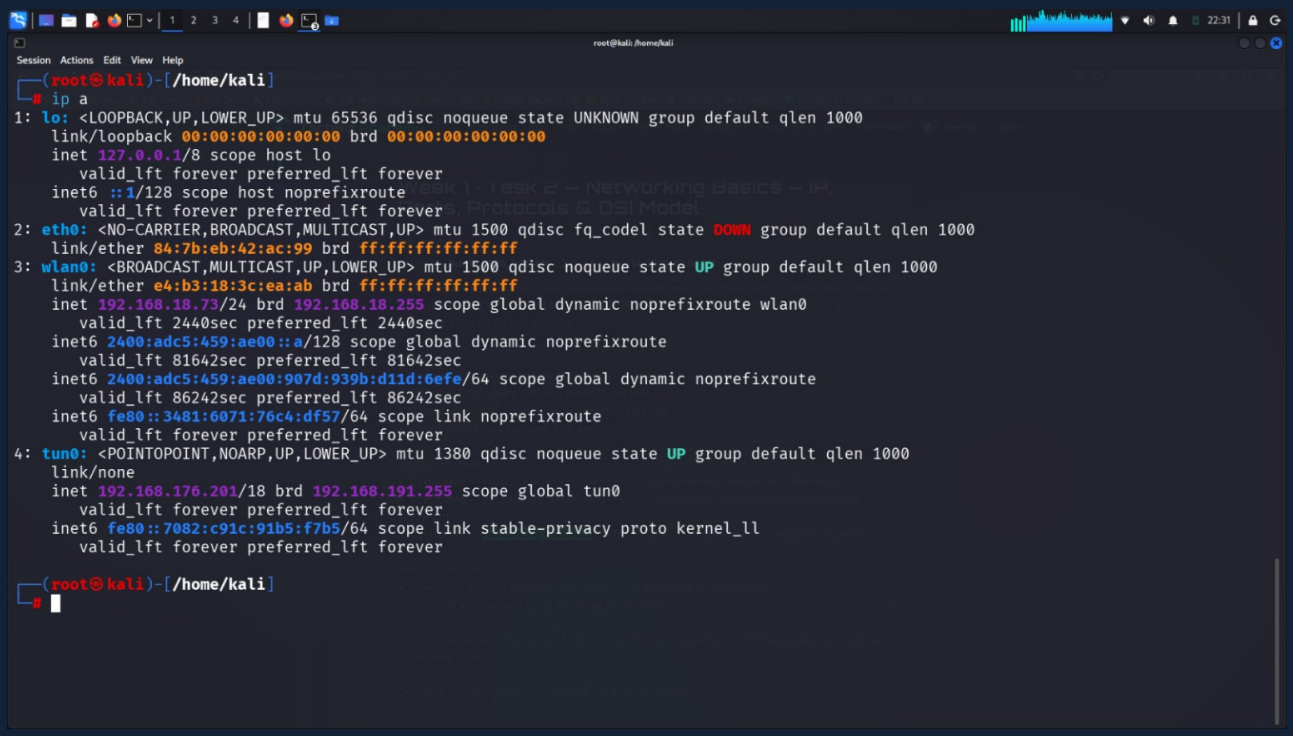



Note: The screenshots show the room completion status as 100%.

# LINUX NETWORK CONFIGURATION

Kali Linux terminal — ip a command output

## PRACTICAL SCREENSHOT

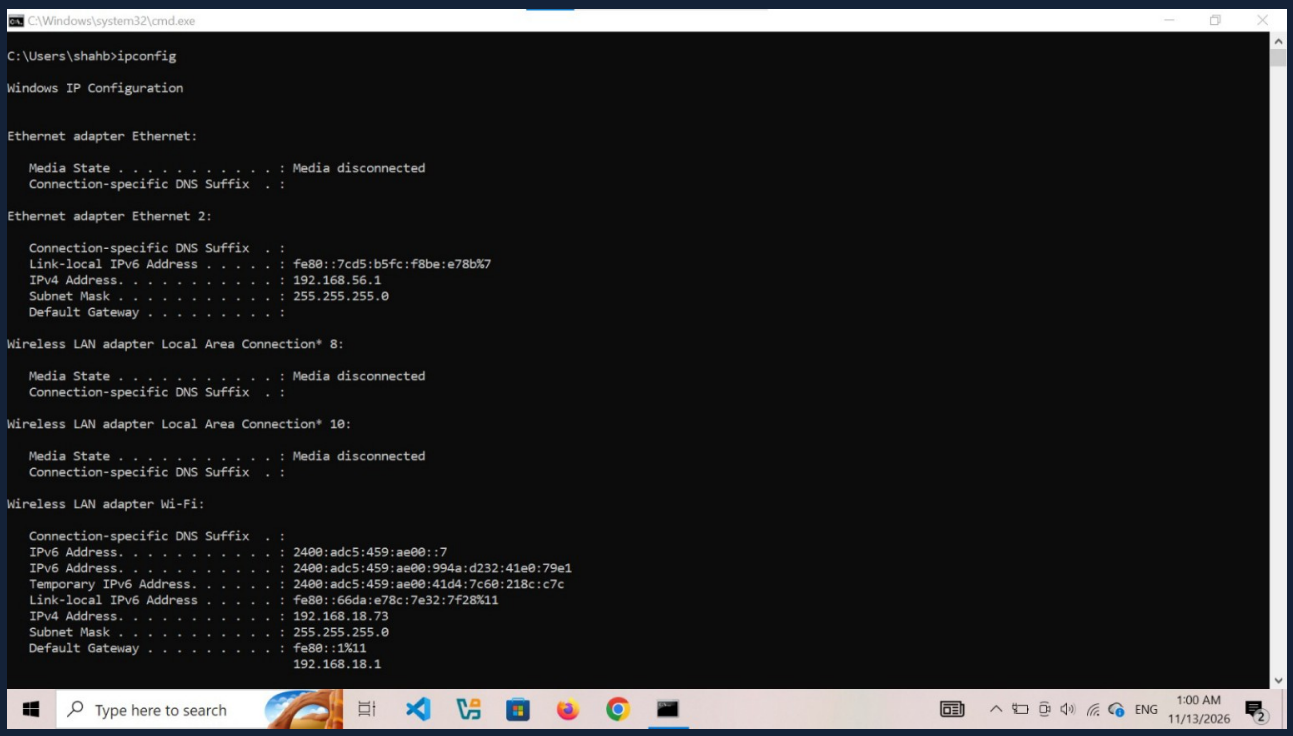



**Detected private IPv4:** **192.168.18.73/24 (wlan0)**

# WINDOWS NETWORK CONFIGURATION

Windows Command Prompt — ipconfig command output

## PRACTICAL SCREENSHOT




**Detected private IPv4:** **192.168.18.73 (Wi-Fi)**